

Examiner reports

Q1.

This integration question proved to be a good start for a large number of students with around 57% correctly choosing option 3, while around 32% chose option 2.

Q2.

This question was generally well attempted with over a quarter of students achieving full marks. Students recognised the need to expand $f'(x)$, integrate their expansion and evaluate the constant of integration from the given condition. Unfortunately the expansion proved to be too demanding algebraically for some, with incorrect squaring of the $2x$ and

$$\left(-\frac{3}{x}\right)$$

terms the major issue.

The integration was generally well done, although some students had issues with the negative power and some omitted the constant of integration which prevented them from completing the solution.

Q3.

This question discriminated well between lower and higher attaining students. The start of the question was accessible to most students. The later parts were more demanding and only the higher attaining students were successful.

Part (a) was completed successfully by a high proportion of students. Many selected a correct approach and manipulated the expression so that the binomial expansion could be applied. The most common error was to extract a factor of 2 instead of $\frac{1}{2}$ from the square root.

Those students who spotted the connection to the first part often did well on part (b). Full marks were available even if the first part of the question had gone wrong, provided $-x^3$ was correctly substituted into their three-term expansion. Sign errors were the most frequently seen mistakes.

In part (c) students could still achieve credit even if their previous answers had gone wrong. Simply substituting their three-term answer to part (b) to form the integral would achieve the first mark.

Very few students demonstrated the insight required for (d)(i) and this question was intended to distinguish the most able students. Those who attempted an answer often talked about the shape of the curve or the use of trapezia. Those who realised that the key was that all the terms were positive often score full marks.

The idea being tested in (d)(ii) was standard, but it was made more difficult by the context. Students were not directed to find the range of validity for their expansion, but had to make the connection for themselves. The first mark could be achieved provided they used their expansion from part (b).

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Both parts of question (d) required explanation. Generally, this was not done well. Students needed to state a key fact which must be true/false and then explain the implications of this.

Q11.

The use of the trapezium rule was well understood with a majority of students presenting a fully correct solution to part (a) with the final answer given to the degree of accuracy requested. Students need to take extra care when transferring correct decimal values from a table into the formula; this was particularly relevant in this series to those values which had leading zeros, namely for $f(4)$ and $f(5)$. The most common method error was to use end values 0 and 4 for x instead of the correct end values 1 and 5. A small minority of

students left their final answer in the form $\frac{694}{1105}$ which resulted in the loss of the final accuracy mark. In part (b)(i), most students were able to integrate one of the terms

correctly but it was not uncommon to see the integral of $x^{-\frac{3}{2}}$ resulting in a positive coefficient or an incorrect numerical simplification of $1/-0.5$. Very few students failed to score the method mark in part (b)(ii) for being able to deal with limits in a correct manner.

Q12.

In part (a), the integration proved difficult for many students who gave wrong coefficients such as $\frac{1}{2}$, 2 or $\frac{1}{1-2}$ and incorrect expressions such as $-\frac{1}{1-2x} e^{1-2x}$. The use of

substitution often did not lead to a correct result, with $\frac{e^u}{u}$ being common. Many students found the elimination of $\ln$ in $e^{-2\ln 2}$ a challenge. Some students mistakenly thought that

changing their result into a decimal number and showing that it approximated $\frac{3}{8}e$ was a sufficient proof.

In part (b), most students gained the first method mark for giving $\frac{du}{dx} = \sec^2 x$ and the B mark for either changing the limits or using the original limits after their integration. The

replacement of dx with $\frac{1}{\sec^2 x} du$ was sometimes seen for the first accuracy mark. The identity $\sec^2 x = 1 + \tan^2 x$ was often seen but replacing $\sec^2 x$ in terms of u often produced $(1 + u)$ rather than $(1 + u^2)$.

Many students thought that they could integrate an expression that was a mixture of u and $\tan x$ terms. A surprising number of able students had difficulty simplifying $(1 + u^2)$

$\sqrt{u}$ to $\left(u^{\frac{1}{2}} + u^{\frac{5}{2}}\right)$.

Q13.

A full spread of marks was seen on this question, with about 20% of the entry gaining full marks, and about another 20% only scoring 2 marks or fewer.

Part (a). Many students attempted this integral by using parts and produced some complicated looking expressions using incorrect integrals. This integral cannot be done by parts; some students did realise this and swapped to the required substitution method. Those who did use an appropriate substitution were usually successful, with an occasional error being made in the coefficient. Few students could do the integral by inspection.

Part (b). Most students knew they had to separate the variables and integrate both sides, with the vast majority using their result from part (a). Most used correct notation in the separation, but some students read e^{2y} as e^2y and some did not multiply across correctly by e^{2y} . Of those who did, most solved the integral of e^{2y} correctly. Those who also solved the integral in x correctly usually went on to complete successfully, although the common error was to think $\ln(a + b) = \ln a + \ln b$. Students who did use their result from part (a), whatever it was, could score method marks for using (1,0) to find a value for their constant, and to use logarithms to express their final answer as $y = f(x)$.

Most students did include a constant, and did attempt to take logs to obtain a final solution.

Q14.

- (a) Those candidates who began by finding $\frac{dy}{dx}$ usually found the correct gradient and most gave the equation of the tangent in the required form. A few made arithmetic errors such as $y - 6 = 9x + 9$ leading to $y = 9x + 3$. A few, having found the gradient to be 9, then incorrectly used the negative reciprocal, i.e. $-\frac{1}{9}$. Some candidates did not realise that differentiation was required and attempted to find the gradient using the coordinates of the two points $(-1, 6)$ and $(2, 33)$.
- (b) (i) Provided no arithmetic errors were made, particularly evaluating $2^5 = 64$, then the correct value of k was found from the equation of the curve. A few candidates used the tangent equation and although they found that $k = 33$, no credit was given.
- (ii) Those who used the equation of the line did not always earn the mark as a concluding statement such as "therefore lies on the tangent" was not made. Some worked backwards and found the gradient using $(2, 33)$. A few candidates confused the requests in parts (i) and (ii) and consequently earned no marks.
- (c) (i) The majority of candidates integrated correctly. Most substituted the limits in the correct order but many sign and arithmetic errors were made. A common error was evaluating $\frac{5}{6} - \frac{54}{6}$ as $\frac{49}{6}$ instead of $-\frac{49}{6}$ and the correct value of the integral was rarely seen.
- (ii) Very few candidates found the area of a trapezium. Most found the area of a triangle instead. Those who used integration in order to find the area under the straight line were far more successful than those who tried to remember the

basic formula from GCSE.

Q15.

This question tested expansion of brackets and integration of negative exponents.

The majority of students used the binomial expansion or expanding brackets successfully in part (a). Typically in a(ii) they repeated their working from part (a)(i), replacing y with x^{-2} , rather than replacing it in their simplified expansion. Most students scored at least M1 for this part, however a significant number of students were unsuccessful in finding a correct expression for $(2 - x^{-2})^3$ and hence did not find the correct values of p and q .

Those students who did not use “hence” but used a correct different method to the one they used in a(i), could gain a maximum of 2 out of the 3 marks. Part (b) made use of the answer to a(ii) and those that had found values for p and q were generally able to correctly integrate their expression, $p + qx^{-4}$, and evaluate it using the given limits in a correct manner.

Q16.

After a successful start with the partial fractions many students went awry in later parts of the question, although fully correct solutions were seen from about 16% of the entry.

- (a) (i) The partial fractions were found correctly by the vast majority of students. Virtually all students knew what to do and had a correct opening line and only the occasional arithmetic error followed in the finding of A and B . The most popular method was to substitute $x = -2$ and $x = \frac{1}{3}$, although some used simultaneous equations and others a mix of the two methods, often choosing to substitute $x = 0$.
- (ii) Again the vast majority of students had the correct form of the integrals but many errors were made in the coefficients, particularly with the $(1 - 3x)$ term, often given simply as $\ln(1 - 3x)$ or $+\frac{1}{3} \ln(1 - 3x)$ and the $(2 + x)$ term was sometimes integrated $\frac{3}{2} \ln(2 + x)$. Most students went on to evaluate their integrals in the correct way, although it was not always clear how they used the given limits and many made a sign error to obtain $3 \ln 2 - \frac{1}{3} \ln 4$, although $\ln 1 = 0$ was well known. Many students who had the definite integral correct could not convert to the form of the requested answer. Several of those who got as far as $3 \ln 2 + \frac{2}{3} \ln 2$ did not see that all they needed to do was add 3 to $\frac{2}{3}$; the answer $\ln 8$ was common. The term rational may not have been sufficiently well known.
- (b) (i) It was expected that students would require little work to find the value of C ; it

is immediate from the algebraic fraction by inspection from $-6x^2 / -3x^2$. However, relatively few noticed this. Many chose long division and found the correct value, but some used inefficient and time consuming methods based on undetermined coefficients. Some found C to a function of x although most gave C as a constant. Some students did not attempt this part of the question.

- (ii) All those who attempted this part knew that the requested area was given by integrating the function from part (b)(i), and most students recognised that this linked the parts of the question together. They gained credit if they had an exact answer to part (a)(ii) and added 2 to it, although many had a sign error when applying the limits and subtracted 2. Some students did not see the connection and tried to do the integral again, this time often going awry with answers such as $\ln(2 - 5x - 3x^2)$.

Q17.

There was a full spread of marks seen in this question with about a quarter of the students scoring 8 or more of the 10 marks available, but a similar proportion scoring 2 or less.

- (a) Most students recognised that this required integration by parts, although some just 'integrated' the t term and the cosine term independently, for no credit. Some tried to use substitution, and indeed some students substituted $u = \frac{\pi t}{4}$ and then used parts. Most identified the parts the right way round but those who did not should have found the integral becoming impossibly harder but they were often determined to get a result, again for no credit. Of those who started correctly, many made a mistake in writing down the parts formula, often confusing the sine and cosine terms. Many had the required coefficient $\frac{4}{\pi}$ inverted. Many who had the coefficients correct at this stage then made an error when integrating the sine term, often getting $\frac{8}{\pi}$ or similar. Many made a mistake when simplifying a correct expression, which was not penalised until part (b).
- (b) Most students separated the variables correctly, and usually used the correct notation. Occasionally the x was dropped or the dx or dt did not appear or was in the wrong place. A few attempted to 'integrate' the expression as given in the question to produce answers of no value. Most gained the mark for integrating x correctly, although some used logarithms due to incorrect separation. Most of the students also gained the method mark for using their result from part (a) in an attempt to find a constant of integration. Some students just substituted $t = 0$, $x = 4$ into the differential equation itself, which made no sense. Those who had the correct form of the integral from part (a) could then score the remaining method mark for calculating firstly x^2 and then x . Many were correct to this point and then misunderstood the request to give their answer to the nearest centimetre or they evaluated using degrees rather than radians. About 8% of the entry gave a fully correct solution.

Q18.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the

gradient m to be -8 . With the answer given, it was not acceptable to just state that the gradient is -8 without some clear indication that this value came from the value of $\frac{dy}{dx}$ when $x = 8$. In part (c), the most common loss of a mark was through failing to simplify $\frac{3}{\left(\frac{8}{3}\right)}$ correctly. The error of integrating $x^{\frac{5}{3}}$ to get a term in the form $kx^{\frac{7}{3}}$ was penalised more heavily.

In part (d), those who found the area of the triangle by solving the equations of OP and AP simultaneously to find the coordinates of P and then using $A = \frac{1}{2}bh$ were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for OP and then 3.2 to 8 for AP . It was also surprising to see a significant minority of candidates finding the lengths of OP and AP and the angle OPA and then using $\frac{1}{2}ab\sin C$ to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

Q19.

In part (a), virtually all candidates showed confidence with finding the partial fractions and, bar a few slips, had this correct. Using two values of x to find A and B was by far the most popular method, although some candidates set up and solved simultaneous equations.

For part (b), there are several ways of finding the coefficients C and D , and these were all seen in candidates' work. The anticipated method of long division was seen less often than multiplying through by the denominator and then using two values of x or equating coefficients of powers of x or a mix of the two. Those who used long division sometimes failed to interpret their result. Using $x = 0$ readily gives the value of D , but those who used

$x = \frac{3}{2}$ in finding C often made a mistake; the value of C is most readily found by equating the coefficients of x^3 . Those candidates who did not use an efficient method here would have wasted time that may have been more gainfully used in other questions.

A significant minority failed to make any progress in part (c) through not seeing the connection to parts (a) and (b). Most immediately went for an incorrect logarithmic integral and scored no further marks. Those who did attempt to use the results from parts (a) and (b) often made sign and / or coefficient errors during the integration. Some candidates who otherwise had this part all correct, failed to score the last mark because they made an error in attempting to manipulate their logarithms to the required form.

Q20.

A substantial proportion of candidates solved this differential equation successfully, which is an impressive achievement given that there is a considerable amount of careful manipulation involved. Most candidates started with a correct separation, with most of

these using correct notation. Many integrated $\frac{1}{y^2}$ correctly, but this was not the case with the integration of $x\sin 3x$. Some showed no detail of an attempt at parts and scored no further marks. Of those who did show stages in their parts integration, some made sign and / or coefficient errors. Most candidates knew that they were to include a constant and attempted to find it, but they needed to have a $\cos 3x$ and a $\sin 3x$ term in their expression

to demonstrate knowledge of $\sin \frac{\pi}{2} = 1$ and $\cos \frac{\pi}{2} = 0$ when finding the value of the constant. The final stage of factoring out a 9 and inverting was done successfully by most who had got this far, although some incorrectly added the constant after inverting and scored no further marks. Those who failed to gain the last mark usually made an error in their last line or two of working, making a sign error or dropping a 3 or even the x .

Q21.

In part (a)(i), the coefficient of x , being an odd number, caused problems for some when completing the square; most students could not square 1.5 without a calculator; those who used fractions could not always simplify $5 - \frac{9}{4}$, and for those using decimals $-1.25 + 5 = 3.75$ was commonly seen.

In part (a)(ii), the equation of the line of symmetry was not usually stated correctly, with several writing down the equation of a curve rather than a straight line; the incorrect equation $y = \frac{3}{2}$ was seen far too often, whereas others simply wrote down numbers such as $\frac{3}{2}$ or $\frac{11}{4}$; this topic was not well understood by many students.

In part (b)(i), most attempts earned the method mark for eliminating y and collecting like terms, but, surprisingly, not all were able to solve the resulting quadratic equation. A common incorrect solution to $x^2 - 4x = 0$ was $x = \pm 2$, whereas others obtained $y = x^2 - 4x$ from the two equations and made no progress.

Integration of polynomials is well drilled and part (b)(ii) earned full marks for weak and strong students alike. There were few errors seen but the most common involved the constant term; some kept the term as 5 and others omitted the final term possibly confusing differentiation with integration.

In part (b)(iii), most students realised the need to substitute appropriate limits but, without a calculator, arithmetic errors were abundant; 4^3 was often evaluated as 48 and the fractions and minus sign caused problems for many students who could not simplify $\frac{64}{3} - 24 + 20$.

Those who were unable to complete part (b)(i) often chose the value 5 as their upper limit, possibly since the coordinates of A were $(0, 5)$, but no credit was given for this. However, those who had the solution $x = 2$ from their earlier work were given a method mark for using this value as their upper limit. A large number of students presented the value of the definite integral as their final answer to the area of the shaded region instead of considering the difference of the area of the appropriate trapezium and the value of the integral.

Q22.

This question, which tested positive rational exponents and integration, was another good source of marks for many students, although for a large minority finding the square of $x^{\frac{3}{2}}$ proved to be more of a challenge than was hoped, with $x^{\frac{9}{4}}$ appearing far too often from otherwise promising students. The indefinite integration and evaluation of a corresponding definite integral were tackled with much more success, with many correct solutions seen.

Q23.

In part (a)(i) of this calculus question, where students were asked to verify that a given expression was equal to zero when $x = 2$, a minority of students failed to give adequate written evidence, particular when the order of calculation was important; the minimum examiners accepted here was $3 \times 4 - \frac{4}{4} - 11 = 0$, although $12 - 1 - 11 = 0$ was more convincing. In part (a)(ii) the differentiation of $-\frac{4}{x^2}$ was a hurdle not always overcome by the weaker students. However, those who first re-wrote it as $-4x^{-2}$ usually went on to differentiate it correctly. Many students scored the mark for giving a correct reason in part (a)(iii). In part (b) a large number of the students did realise that they needed to integrate the given expression for the gradient to find the equation of the curve but a significant number of them did not include the arbitrary constant which is essential here, and only a minority knew how to find its value from the given information. Although only a minority, it was still disappointing to see more students than expected writing down the equation of the curve in the form $y - y_1 = m(x - x_1)$.

Q24.

In part (a)(i) candidates who recognised that the first step required was differentiation generally produced correct answers, but a few had trouble with the “ $-x$ ” term and others were unable to combine $-1 + (-4)$ correctly. Some weaker candidates assumed that they could find the gradient using the coordinates of two points.

In part (a)(ii) once again there was much confusion between parallel and perpendicular lines, with many candidates giving the wrong gradient of the tan gent. To earn the final mark it was essential to express the equation in the specified form and some failed to do so; for instance, an answer such as $y + 5x = 17$ is not in the correct form.

In part (b)(i) the integration was generally correct and most candidates made an attempt to substitute the correct limits. A few however could not integrate “14” and left it as a constant. Very few were able to combine their fractions correctly and consequently only a small proportion of the candidates found the correct value of the integral.

In part (b)(ii) candidates who had only found the definite integral in part (b)(i) were generously rewarded with the remainder of the marks here. Not everyone recognised the need to find the area of a triangle in order to find the area of the shaded region. A few correctly found the area under the line $y = 4x + 8$ by integration, but this is not the recommended approach.

Q25.

A majority of candidates gave the correct binomial expansion in part (a), although a common error amongst the less successful candidates was to write the third term as $-3x^2$ instead of $+3x^2$. In part (b), most candidates picked up the method mark for their expansion of $(1+y)^4$. However, it was not uncommon to see errors in collecting like terms, the most common of which were ' $6y^2 - 3y^2 = -3y^2$ ' or ' $6y^2 - 3y^2 = 9y^2$ '. It was disappointing to see a small minority of candidates either having expansions entirely in terms of x or having a mixture of terms involving x and y with no indication of any corrections despite the printed answer.

Part (c) was generally only answered correctly by the most able candidates who replaced y in their expansion for part (b) by $x^{0.5}$ and then integrated the resulting expression with respect to x . A common error amongst other solutions was to integrate the expansion from part (b) with respect to y , with some either leaving their answer in terms of y or others replacing y after the integration by $x^{0.5}$.

Candidates who didn't appreciate the link between the parts and expanded again often met problems with either the 'new' expansions or the powers involved.

Q26.

Most candidates replaced $\frac{8}{x^4}$ by $8x^{-4}$ and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a '+c' in their

answer. A very large majority of candidates realised that the value of $\frac{dy}{dx}$ at $x = 1$ was required in

part (b), but a small minority did not appreciate that the value of y was also required. Although many candidates found the equation of the tan gent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for $\frac{dy}{dx}$ to zero and attempted to solve the resulting equation, although only about half of the candidates obtained the correct coordinates of M . Those who had wrong powers of x in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving

$\frac{d^2y}{dx^2} > 0$ or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate $\frac{8}{x^4}$ correctly, but errors in the integration of $(x + 3)$ or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to

part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the 'Hence', showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the x -axis being a tangent), 12 (the y value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

Q27.

Part (a) was very well answered, with the majority of candidates obtaining the correct function. The main error made by candidates who obtained a $\ln$ function was with the multiplying constant, which was often 2 or was completely missing. Omission of brackets was also a problem.

Not many of the candidates answered part (b) fully correctly. Only a few integrated by parts 'the

wrong way round' and some weaker candidates put $u = x$ and $v = \sin \frac{x}{2}$. However most knew

that they should integrate the $\sin \frac{x}{2}$ term, but did it wrongly, getting $\pm \cos \frac{x}{2}$ or more commonly

$\pm \frac{1}{2} \cos \frac{x}{2}$. If the first method mark was earned, the second was generally scored as well.

Q28.

A full range of marks were seen in this question. Some candidates had little or no idea as to how to solve a differential equation, whereas some gave concise and correct solutions and could use their solution to solve the problem in the given context.

In part (a)(i) most candidates separated the variables correctly, although some poor

notation was seen and some mishandled the $\sqrt{x}$. Many also made errors in the integration, often getting the coefficient wrong in one or both integrals. Although many did add an arbitrary constant, many also failed to complete the question as requested and by writing x in terms of t . Of those that did, many failed to square their expression in t correctly; $(f(t) + C)^2 = (f(t))^2 + C^2$ was a common error. Many apparently didn't review their solution when given the form of answer in part (a)(ii).

When manipulating an expression containing a constant some, for example, renamed

$\frac{C}{2}$ as C

and sometimes then confused themselves; whereas others correctly introduced a new letter for the revised constant.

In (a)(ii) many went on to find a correct value for the constant and give the required form,

as they used a previous line of working from part (a)(i) rather than their final answer. Any attempt to find a constant using the information given was credited, but many tried to force their solution into the form of the answer given, rather than review their work for an error.

In part (b)(i) the question confused a lot of candidates and many assumed the greatest height occurred at $\cos bt = 0$ and not -1 . Many candidates didn't attempt this part.

In (b)(ii) for those who set up a correct equation, the most common error here was to use degrees instead of radians; so the more common answer was 207.3, it apparently not occurring to candidates that a time of over 3 minutes was rather long for a fairground car to reach a height of 5 m. Many did not set up a correct form of the equation and so could gain no marks.

Q29.

In part (a) most candidates were well drilled in integration and earned the first three marks; however, the fractions caused tremendous problems to most candidates who were simply

incapable of evaluating $\left(\frac{1}{4} - \frac{2}{3} + 3\right) - \left(\frac{1}{4} + \frac{2}{3} - 3\right)$ without a calculator. Consequently, an incredibly small percentage obtained the correct value for the integral.

In (b) the area of the triangle was usually found to have value 2, although quite a few could not

simplify $\frac{2 \times 2}{2}$, and most candidates scored a mark for subtracting the area of the triangle from their answer to part (a). Only the best candidates found the correct value for the shaded region.

Q30.

The majority of candidates answered part (a) successfully using the binomial expansion, Pascal's triangle or expanding brackets. Those candidates who wrote the expansion using the binomial coefficient notation from the formulae book did not always convince the examiner that they knew how to correctly evaluate the combinatorial coefficients.

In part (b)(i) very few candidates scored full marks with a large number of attempts seen where the numerator and denominator were integrated separately. Those candidates who realized the need for negative indices generally scored well. Most candidates picked up the method mark in the final part of the question for dealing with the limits in a correct manner. A few candidates just stated the correct answer for the definite integral without showing any working. Such an approach gained no credit as it did not explicitly indicate the use of 'Hence'.

Q31.

Most candidates scored the marks for the two derivatives with relatively few omitting the minus signs. Some candidates managed to drop the 2 from $\cos 2\theta$. Many candidates brought the

commonly used parameter t into their expression using $\frac{dx}{dt} = -\sin\theta$ or similar, but were not penalised. Virtually all knew how to use the chain rule, but many got mixed up with the coefficients in both expanding $\sin 2\theta$ and simplifying the expression. The common error was to find $k = 3$. Some candidates got a result in terms of $\sin \theta$ and some got no result at all, but any value of k , however found or invented, was allowed in part (a)(ii).

Part (a)(ii) was generally done well with most candidates demonstrating they both knew and could use the relation between the gradients of the tan gent and the normal, although a few

found an equation of the tan gent. The evaluation using $\theta = \frac{\pi}{3}$ was generally accurately done; however candidates had to have derived $k = 6$ correctly to gain the final mark for the equation of the normal.

In part (b) it was expected here that candidates would note the whole question was about the use of double angle formulae, and many did. Most got as far as at least trying to express $\sin^2 x$ in term of $\cos 2x$ and of those, many did it correctly, even if sometimes there was a mix of variables. The integration of $\cos 2x$ was usually accurate, with the common error being to double $\sin 2x$ rather than halve it. There were few sign errors. Virtually all candidates who achieved an

integral knew how to use the limits, but the real test here was in handling $\sin 2\left(-\frac{\pi}{4}\right)$; many candidates lost credit through not showing their evaluation or getting the sign wrong.

However, many candidates, about 25% of the entry, went on to obtain the correct answer and give it in an exact form, although some did make a mistake with the signs, particularly the three minus signs involved with the lower limit. Of the few candidates who attempted the integral using parts most made little progress beyond the first line of working, not realising that in order to proceed with the integral they needed to change $\cos^2 x$ back to $\sin^2 x$, and manipulate the resulting expression.

Q32.

Most candidates answered parts (a) and (b) correctly; the most common incorrect values

for k in part (a) were either $5\frac{1}{2}$ or $\frac{1}{5}$, and in part (b) it was disappointing to see 4 being integrated as $\frac{4^2}{2}$. The final part of the question was generally answered very badly with the incorrect answer, $y = 3x$, the equation of a line, seemingly appearing as often, if not more so, than the correct equation of the curve.

Q33.

In part (a), most candidates knew to multiply through by the denominator and many did this correctly and went on to find A and B successfully. However, some candidates did not apparently know how to handle the 2, some omitting it altogether and some including it but not multiplying it by the denominator. Some candidates multiplied out the denominator and divided by the resulting quadratic expression to separate out the 2, then worked with the

resulting algebraic fraction. Although this was unnecessary, it was a perfectly acceptable method. Relatively few candidates used the method of setting up simultaneous equations to find A and B .

In part (b), virtually all candidates knew they were to use the partial fractions to do the integration, and most recognised that log integrals were involved. The common error was

to miss the multiplier $\frac{1}{5}$ when integrating $(5x - 1)^{-1}$. Several candidates inverted the coefficient, giving it as $\frac{5}{7}$. Although most candidates integrated 2 to $2x$, many omitted the arbitrary constant.

Q34.

Marks for this question covered the whole range from 0 to 6. Many candidates knew the topic well and gave a concise and correct solution for full marks. Most candidates separated the variables correctly with conventional notation, the common error being to

have $\frac{1}{y}$ on the left hand side followed by a log integral; there were also many errors in integrating. Although most candidates knew this involved sine, there were errors with signs

and the coefficient was often given as $\frac{1}{3}$ rather than 3. Some candidates dropped the 3 completely.

Candidates who had only made sign or coefficient errors could gain partial credit when finding a constant of integration and many did. Pleasingly, virtually all candidates who attempted this worked in radians. Some candidates omitted the constant altogether and substituted the given values into their solution, or even the given differential equation to produce nonsense. Some candidates confused themselves by multiplying by 2 before finding the value of their constant and then halved it again at the end.

Q35.

The sign of p was often awry, although that of q was usually correct. This was not penalised again in part (b)(i) where a lack of simplifying the coefficients and signs in the expressions was a greater source of mark loss. It was also disappointing to see a

significant minority of candidates integrating the term $+\frac{1}{x^6}$ instead of the printed term $-\frac{1}{x^6}$. However, in general, most candidates showed that they understood how to integrate x raised to a negative power.

Most knew how to evaluate a definite integral in part (b)(ii) but very few could maintain the numerical accuracy needed for the final mark. Those few candidates who just wrote down the value of the definite integral, presumably from a calculator, without showing how it had been obtained from the previous part (Hence) were awarded no credit. Some candidates, having correctly obtained -1.7 for the value of the definite integral, went on to write 'Area = 1.7' as their final line of solution. This was not penalised and such candidates were awarded both marks.